



NORTH SYDNEY BOYS HIGH SCHOOL

2017 HSC ASSESSMENT TASK 1

Mathematics

Extension 1

General Instructions

- Working time – 60 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

Student Name/Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total %
Mark	$\overline{13}$	$\overline{3}$	$\overline{11}$	$\overline{13}$	$\overline{40}$	$\overline{100}$

Question 1 – Integration (13 Marks)

(a) Find:

(i) $\int (5x + 2)dx$ **1**

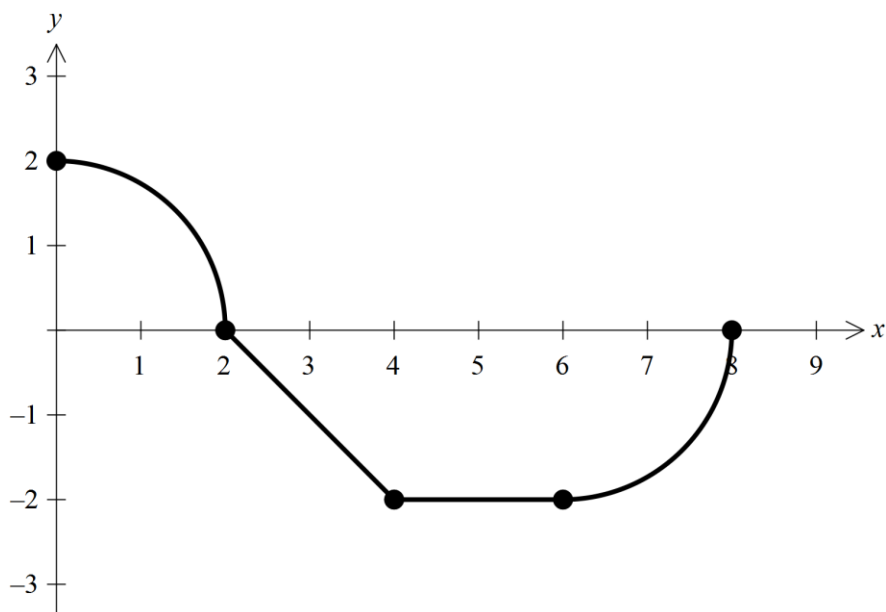
(ii) $\int \frac{x^3 + 1}{x^2} dx$ **2**

(b) Evaluate:

(i) $\int_{-1}^1 (x^3 + 1) dx$ **2**

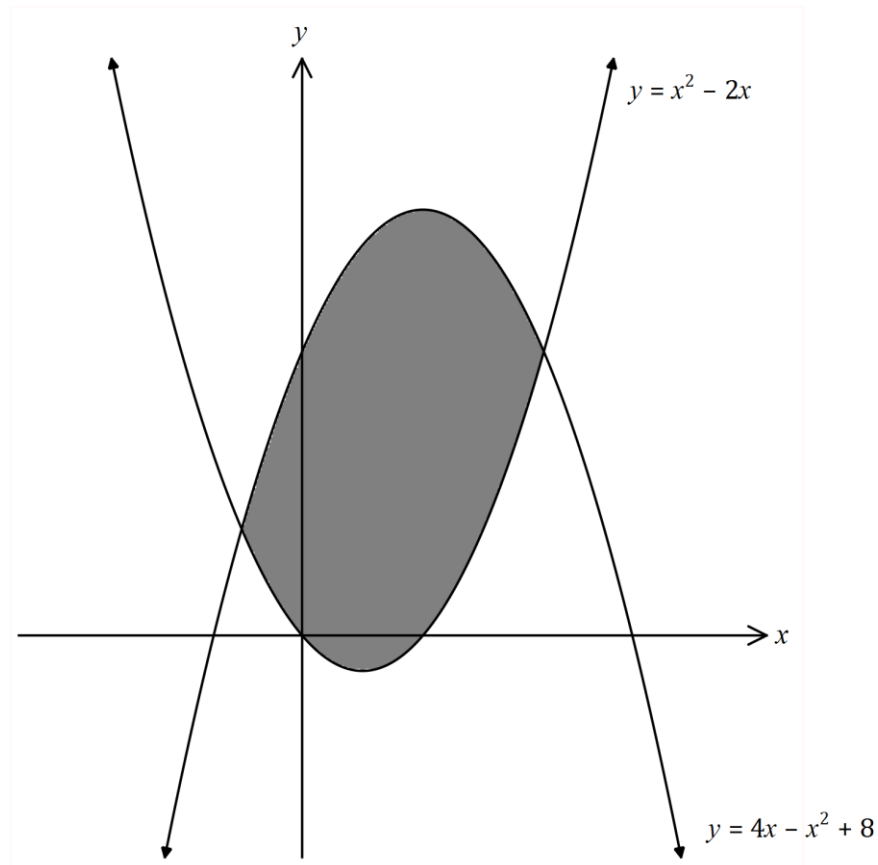
(ii) $\int_1^4 \frac{dx}{\sqrt{x}}$ **2**

(c)

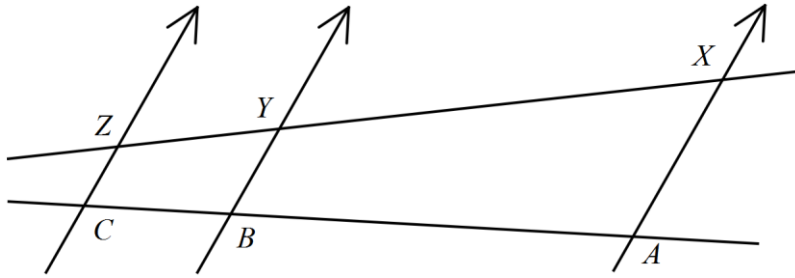


With reference to the above diagram, evaluate $\int_0^8 f(x)dx$ **2**

- (d) (i) Find the coordinates where $y = 4x - x^2 + 8$ and $y = x^2 - 2x$ intersect. **2**
- (ii) Calculate the area bounded by the two curves shown below. **2**



Question 2 – Euclidean Geometry (3 Marks)



In the diagram ABC and XYZ are two transversals which cross a family of the parallel lines. $AB = 3BC$ and $XY = YZ + 3$, where all the measurements are in centimetres.

(i) State a reason why $\frac{AB}{BC} = \frac{XY}{YZ}$.

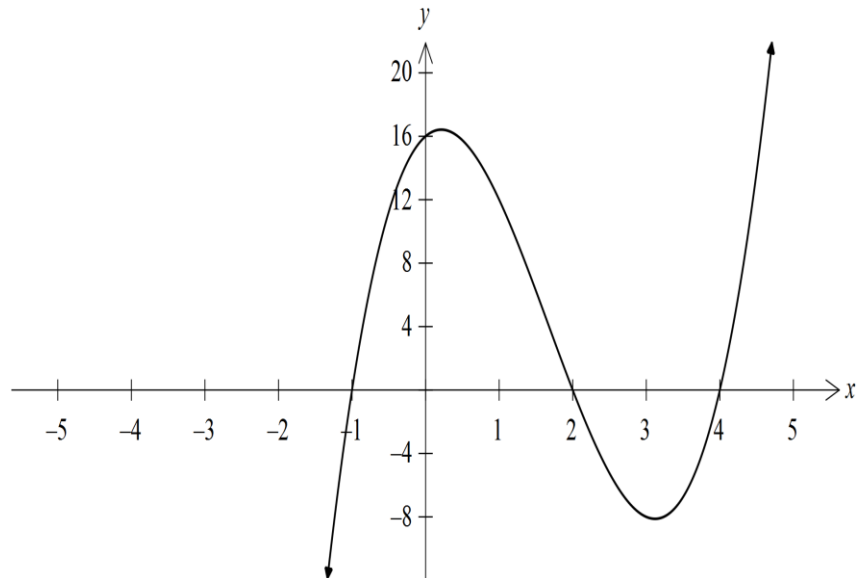
1

(ii) Hence find the lengths of XY and YZ .

2

Question 3 – Polynomials (11 Marks)

(a)



(i) Find the equation of the polynomial shown above.

2

(ii) With reference to the above polynomial or otherwise, solve the following:

$$x^3 > 5x^2 - 2x - 8$$

2

(b) $P(x) = x^3 + ax^2 + 2x + b$ has a factor of $(x + 3)$ and a remainder of 4 when it is divided by $(x - 1)$.

3

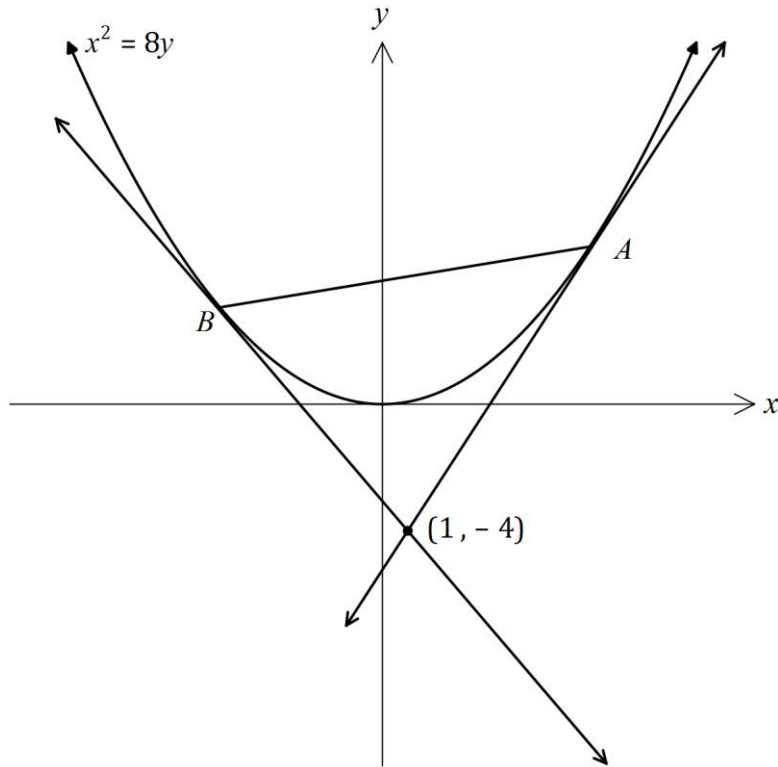
Find the values of a and b .

(c) Solve the equation $8x^3 - 14x^2 - 7x + 6 = 0$ given that two of the roots are reciprocals of each other.

4

Question 4 – Parametrics (13 Marks)

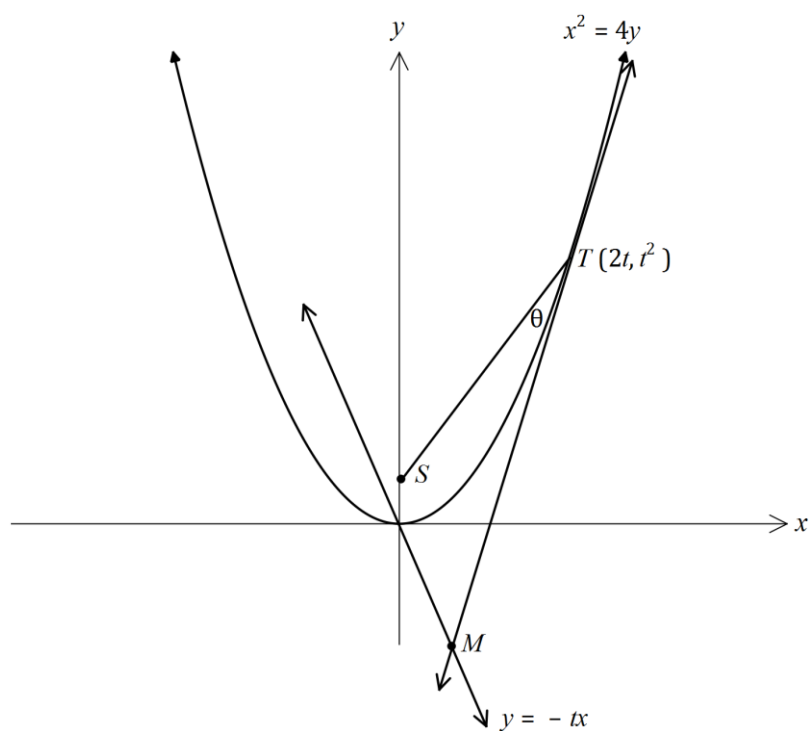
- (a) If $x = 1 + \cos \theta$ and $y = 2 - \sin \theta$ find a relationship between x and y only. **2**
- (b)



The diagram above shows the parabola $x^2 = 8y$. Tangents are drawn from points A and B on the parabola such that they intersect at $(1, -4)$. Find the equation of AB . **2**

- (c) The horizontal line $y = k$, where $k > 0$, cuts the parabola $x^2 = 4ay$ at the points $P(2at, at^2)$ and $Q(-2at, at^2)$. If O is the origin and $\triangle OPQ$ is equilateral, express k in terms of a . **3**

(d)



The parabola $x^2 = 4y$, with focus S , has a tangent at $T(2t, t^2)$.

The acute angle θ is the angle between ST and the tangent at T .

(i) Show that the tangent at T has a gradient of t . **1**

(ii) The tangent at T intersects the line $y = -tx$ at M . Find the locus of M as T moves on the parabola. **3**

(iii) Find $\tan \theta$ in terms of t . **2**

End of examination

Question 1

$$(a) (i) \int (5x + 2) dx = \frac{5x^2}{2} + 2x + C$$

(a) (ii)

$$\begin{aligned} \int \frac{x^3 + 1}{x^2} dx &= \int (x + x^{-2}) dx \\ &= \frac{x^2}{2} - \frac{1}{x} + C \end{aligned}$$

(b)(i)

$$\begin{aligned} \int_{-1}^1 (x^3 + 1) dx &= \left[\frac{x^4}{4} + x \right]_{-1}^1 \\ &= \left(\frac{1^4}{4} + 1 \right) - \left(\frac{(-1)^4}{4} - 1 \right) \\ &= 2 \end{aligned}$$

(b)(ii)

$$\begin{aligned} \int_1^4 \frac{dx}{\sqrt{x}} &= \int_1^4 x^{-\frac{1}{2}} dx \\ &= \left[2x^{\frac{1}{2}} \right]_1^4 \\ &= 2\sqrt{4} - 2\sqrt{1} \\ &= 2 \end{aligned}$$

(c)

$$\begin{aligned} \int_0^8 f(x) dx &= \left(\frac{1}{4} \right) (\pi)(2^2) - \left(\frac{1}{2} \right) (2)(2) - (2)(2) - \left(\frac{1}{4} \right) (\pi)(2^2) \\ &= -6 \end{aligned}$$

(d) (i)

$$4x - x^2 + 8 = x^2 - 2x$$

$$2x^2 - 6x - 8 = 0$$

$$2(x - 4)(x + 1) = 0$$

$$x = -1 \text{ or } x = 4$$

∴ intersections are (-1, 3) and (4, 8)

Question 1 – Continued

$$\begin{aligned} A &= \int_{-1}^4 ((4x - x^2 + 8) - (x^2 - 2x)) dx \\ &= \int_{-1}^4 (-2x^2 + 6x + 8) dx \\ &= \left[\frac{-2x^3}{3} + 3x^2 + 8x \right]_{-1}^4 \\ &= \left(\frac{-2(4)^3}{3} + 3(4)^2 + 8(4) \right) - \left(\frac{-2(-1)^3}{3} + 3(-1)^2 + 8(-1) \right) \\ &= \frac{112}{3} - \left(-\frac{13}{3} \right) \\ &= \frac{125}{3} \text{ units}^2 \end{aligned}$$

Question 2

(i) Parallel lines preserve ratios of intercepts on transversals.

(ii)

$$\frac{AB}{BC} = \frac{XY}{YZ} \text{ (as above)}$$

$$\frac{3BC}{BC} = \frac{YZ + 3}{YZ}$$

$$3 = 1 + \frac{3}{YZ}$$

$$2 = \frac{3}{YZ}$$

$$YZ = \frac{3}{2} \text{ cm}$$

$$XY = \frac{3}{2} + 3$$

$$= \frac{9}{2} \text{ cm}$$

Question 3

(a)(i)

$$y = 2(x+1)(x-2)(x-4)$$

(a)(ii)

$$\begin{aligned}x^3 &> 5x^2 - 2x - 8 \\x^3 - 5x^2 + 2x + 8 &> 0 \\(x+1)(x-2)(x-4) &> 0\end{aligned}$$

$$\therefore -1 < x < 2 \text{ or } x > 4$$

(b)

$$\begin{aligned}P(-3) &= (-3)^3 + a(-3)^2 + 2(-3) + b = 0 \\9a + b - 33 &= 0 \\b &= 33 - 9a \dots \textcircled{1}\end{aligned}$$

$$\begin{aligned}P(1) &= (1)^3 + a(1)^2 + 2(1) + b = 4 \\a + b + 3 &= 4 \\a + b &= 1 \dots \textcircled{2}\end{aligned}$$

$$\textcircled{1} \rightarrow \textcircled{2}$$

$$\begin{aligned}a + 33 - 9a &= 1 \\8a &= 32 \\a &= 4\end{aligned}$$

$$\rightarrow \textcircled{1}$$

$$\begin{aligned}b &= 33 - 9(4) \\b &= -3\end{aligned}$$

Question 3 - continued

(c)

let the roots be α , $\frac{1}{\alpha}$ & β

$$\cancel{\alpha} \times \frac{\cancel{1}}{\cancel{\alpha}} \times \beta = -\frac{6}{8}$$

$$\beta = -\frac{3}{4}$$

$$\alpha + \frac{1}{\alpha} - \frac{3}{4} = \frac{14}{8}$$

$$\alpha + \frac{1}{\alpha} = \frac{20}{8}$$

$$\frac{\alpha^2 + 1}{\alpha} = \frac{5}{2}$$

$$2\alpha^2 + 2 = 5\alpha$$

$$2\alpha^2 - 5\alpha + 2 = 0$$

$$(\alpha - 2)(2\alpha - 1) = 0$$

$$\alpha = 2 \text{ or } \alpha = \frac{1}{2}$$

$\therefore$ the roots are $-\frac{3}{4}$, $\frac{1}{2}$ & 2

Question 4

(a)

$$x = 1 + \cos \theta$$

$$\cos \theta = x - 1$$

$$y = 2 - \sin \theta$$

$$\sin \theta = 2 - y$$

$$(x - 1)^2 + (2 - y)^2 = \cos^2 \theta + \sin^2 \theta$$

$$(x - 1)^2 + (y - 2)^2 = 1$$

(b)

$$x x_0 = 2a(y + y_0)$$

$$x = 2(2)(y - 4)$$

$$x = 4y - 16$$

$$x - 4y + 16 = 0$$

(c)

PO = PQ (sides of equilateral triangle)

$$\sqrt{(2at)^2 + (at^2)^2} = 4at$$

$$4a^2 t^2 + a^2 t^4 = 16a^2 t^2$$

$$a^2 t^4 = 12a^2 t^2$$

$$t^2 = 12$$

$$k = at^2 \text{ (since } k = y)$$

$$\therefore k = 12a$$

(d)(i)

$$x^2 = 4y$$

$$y = \frac{x^2}{4}$$

$$y' = \frac{2x}{4}$$

when $x = 2t$

$$y' = \frac{2(2t)}{4}$$

$$= t$$

Question 4 - continued

(d) (ii)

$$tx - (1)t^2 = -tx$$

$$2tx = t^2$$

$$t = 2x$$

$$y = -tx$$

$$y = -(2x)(x)$$

$$y = -2x^2$$

this is a concave down parabola with

a vertex at (0,0) and a focal length of $\frac{1}{8}$

(d) (iii)

$$m_{TS} = \frac{t^2 - 1}{2t - 0}$$

$$m_{TM} = t$$

$$\tan \theta = \left| \frac{m_{TS} - m_{TM}}{1 + m_{TS} m_{TM}} \right|$$

$$= \left| \frac{t - \frac{t^2 - 1}{2t}}{1 + (t) \left(\frac{t^2 - 1}{2t} \right)} \right|$$

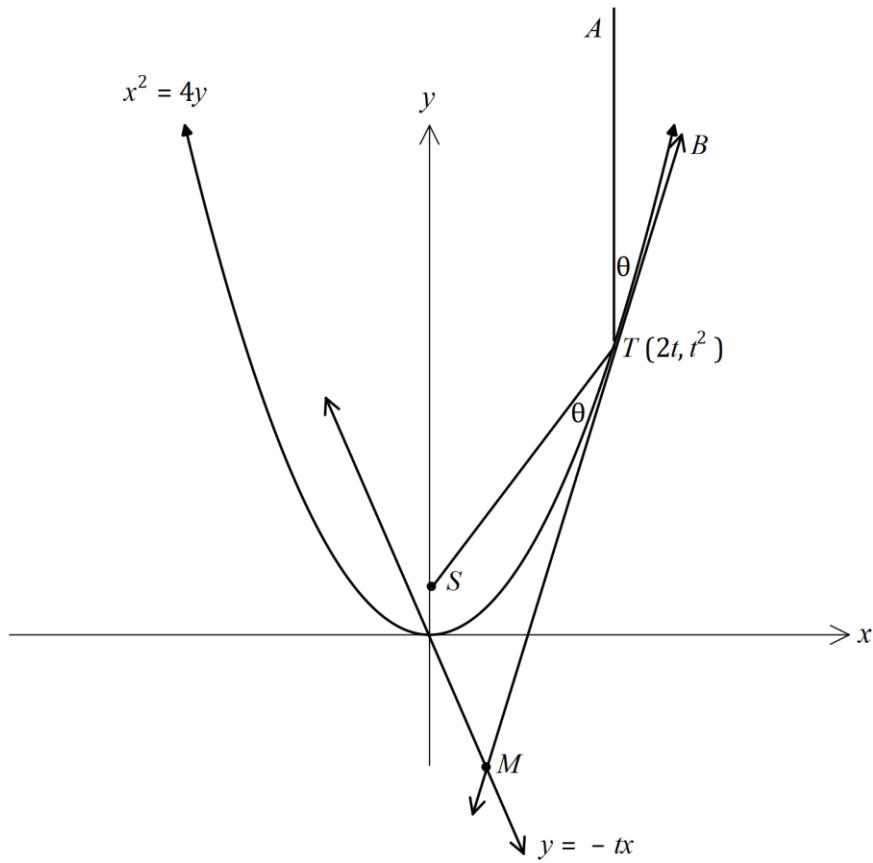
$$= \left| \frac{\left(\frac{2t^2 - (t^2 - 1)}{2t} \right)}{\left(\frac{2t + t^3 - t}{2t} \right)} \right|$$

$$= \left| \frac{t^2 + 1}{t^3 + t} \right|$$

$$= \left| \frac{t^2 + 1}{t(t^2 + 1)} \right|$$

$$= \left| \frac{1}{t} \right|$$

Alternative solution to 4 (d) (iii)



Let point A be as shown such that AT is parallel to the axis of the parabola

Let point B be a point on the tangent at T as shown

$\angle ATB = \theta$ (reflection property of the parabola)

$$\therefore \tan \theta = \frac{1}{t}$$

but since θ is acute

$$\tan \theta = \left| \frac{1}{t} \right|$$